THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · HARD LEVEL UNIT TEST SET 3

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The value of $24^2 - 23^2$ is: 1

(a) 24 **(b)** 47 **(c)** 49 **(d)** 48

2. The sides of two squares are in the ratio 2 : 3. The ratio of their areas is: 1

(a) 4 : 3 **(b)** 2 : 3 **(c)** 4 : 9 **(d)** 2 : 9

3. If the difference between the CI and the SI on a sum for 2 years at 4% is ₹ 12, the sum (in ₹) is: 1

(a) 6750 **(b)** 15000 **(c)** 7500 **(d)** 8250

4. The longest rod that fits in a room 2 m by 3 m by 6 m has length (in m): 1

(a) 6 **(b)** 7 **(c)** 8 **(d)** 11

5. The wages of 12 workers for 20 days are ₹ 36000. The wages of 9 workers for 14 days at the same daily rate (in ₹) are: 1

(a) 36000 **(b)** 18900 **(c)** 27000 **(d)** 19050

6. $(a+b)^2 = a^2 + b^2$ is true only when: 1

(a) $a = 1$ **(b)** $ab = 0$ **(c)** $a = b$ **(d)** $a + b = 0$

7. The number x for which 4 : 9 = 12 : x is: 1

(a) 18 **(b)** 25 **(c)** 21 **(d)** 27

8. A sum triples in 4 years at simple interest. The rate of interest per annum is: 1

(a) 55% **(b)** 50% **(c)** 100% **(d)** 45%

9. The legs of a right triangle are in the ratio $8 : 15$ and its hypotenuse is 51 cm. Its perimeter (in cm) is: 1

(a) 69 **(b)** 120 **(c)** 123 **(d)** 102

10. The present ages of Ria and Sia are in the ratio 2:5. After 9 years the ratio will be 7:13. Ria's present age (years) is: 1

(a) 21 **(b)** 12 **(c)** 10 **(d)** 30

11. If $x + y = 11$ and $xy = 28$, then $x^2 + y^2$ equals: 1

(a) 65 **(b)** 121 **(c)** 93 **(d)** 177

12. If x : y = 3 : 4 and y : z = 6 : 5, then x : y : z is: 1

(a) 3 : 10 : 5 **(b)** 18 : 24 : 20 **(c)** 9 : 12 : 10 **(d)** 3 : 4 : 5

13. Two articles are sold at ₹ 2400 each, one at a gain of 20% and the other at a loss of 20%. The overall result is a loss of (in %): 1

(a) 20% **(b)** 4% **(c)** 0% **(d)** 10%

14. The sides of a rectangle are in the ratio $5 : 12$ and its diagonal is 39 cm. The area of the rectangle (in cm²) is: 1

(a) 102 **(b)** 117 **(c)** 540 **(d)** 549

15. Pipe A fills a tank in 15 hours and pipe B in 10 hours. Together they fill it in (hours): 1

(a) 25 **(b)** 7 **(c)** 6 **(d)** 12.5

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. If $(x + p)(x + q) = x^2 + 8x + 15$, find $p^2 + q^2$. 2

17. Divide ₹ 1200 into parts proportional to $\frac{1}{2}$ and $\frac{1}{3}$. 2

18. Rahul invests ₹ 2500 at 20% p.a. compounded annually. Find the amount after 2 years and write it as a fraction of the principal. 2

19. In a rectangle ABCD, P is a point inside with PA = 8 cm, PB = 6 cm, PC = 10 cm. Find PD^2. 2

20. The present ages of a father and son are in the ratio 7:2. After 6 years the ratio will be 8:3. Find their present ages. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. In a pie chart, the sector for Science is 40° and represents 12 students, while the sector for Maths is 80°. Find the total number of students and the number who like Maths. 3

22. If $x - y = 6$ and $xy = 40$, find $x^2 + y^2$ and $(x + y)^2$. 3

23. The ratio of the present ages of a father and son is 5 : 2. After 6 years the ratio will be 2 : 1. Find their present ages. 3

24. A merchant sells two items at ₹ 2640 each, one at a gain of 20% and the other at a loss of 20%. Find the overall gain or loss in rupees. 3

OR

A ladder 53 m long, with its foot in a street, reaches a window 45 m high on one side. Without moving its foot it is turned to reach a window 28 m high on the opposite side. Find the width of the street. 3

25. A garrison of 180 men has provisions for 40 days. After 10 days 45 men leave. For how many more days will the remaining provisions last for the men who stay? 3

OR

Simplify $(4x + 5)^2 - (4x - 5)^2$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A room is 1 m long, 2 m wide and 2 m high. Find the length of the longest rod that can be placed in it. Also find the diagonal of its floor. 5

OR

24 workers can finish a job in 30 days. After 10 days 8 workers leave. In how many total days will the job be finished? By how many days is it delayed? 5

27. The sides of a rectangle are $(4x + 3)$ m and $(4x - 3)$ m. Find its area as an expression, its value at $x = 6$, and by how much it is less than the area of a square of side $4x$ m. 5

OR

A, B and C invest ₹ 3000, ₹ 2000 and ₹ 4000 for 4, 6 and 2 months respectively. A profit of ₹ 1200 is shared in the ratio of (investment $\times$ time). Find the ratio and each share. 5

END OF QUESTION PAPER **Total: 50 Marks**

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